

Planetary Exploration Rover

Problem Statement

This project is about implementing a toy Planetary Exploration Rover. The project has two parts. This assignment describes the first part. The second part will be assigned to you after the submission of the first part. The dates will be announced as the course proceeds.

Stage 1: Reactive–Deliberative Control

Objective: Build the rover’s core modules

A rover on Mars senses terrain type (flat, rocky, hazardous) and battery level. Based on its precepts, it chooses actions to explore or recharge. Enhance the rover with an A* planner to navigate between waypoints (landmarks, recharge stations, and sample-collection zones). Movement costs battery with the varying terrains costing varying amounts: flat = 5, sandy = 10, radiation-spot = 15, rocky = 1000. For the sake of simplicity of the implementation, assume that the rover moves over a 2-D grid. Each of the tiles of the grid can be flat, sandy or rocky as shown in Figure 1.

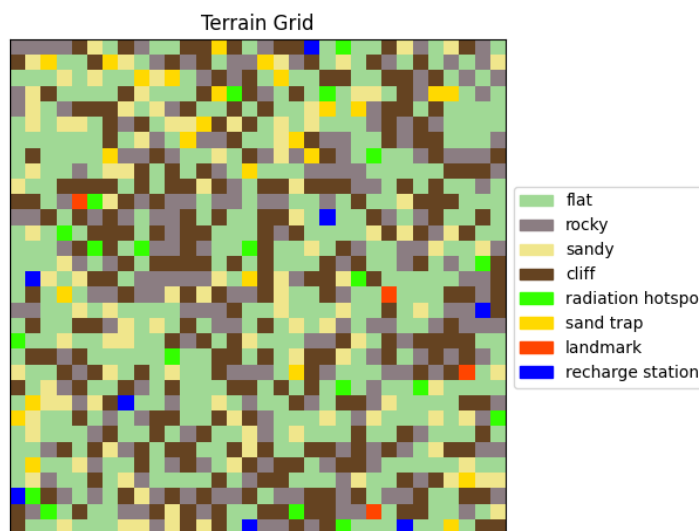


Figure 1: Sample Grid

Rules

- Initiate recharge when battery $< 20\%$.
- Avoid impassable obstacles (rocks).

- If terrain == "hazardous" (e.g., sand trap or radiation hotspot) at the current cell, execute "stop", then backtrack to the last known safe cell.
- If battery between 20%–25% and a recharge station is within two moves (Euclidean distance ≤ 2), override exploration goals and drive directly to recharge station.

Your task is to design a reflex agent with navigation capabilities for the rover. Set an initial point and a distant enough goal. The rover should navigate safely while charging as stated in these rules.

Tasks

- **Task 1:** Implement the reflex module and path planning module as described.
- **Task 2:** Design, implement, and compare 4 different heuristics for A*.

Deliverables

- **Deliverable 1:** Code with your implementation of the aforementioned tasks.
- **Deliverable 2:** A comparison code file (preferably a Jupyter Notebook) comparing the heuristics for A*.
- **Deliverable 3:** Provide a short ($\tilde{2}$ pages) progress report describing your implementation of the reflex and path planning module. Explain the 4 heuristics you have implemented for A* and provide a descriptive comparison report.

Deadline: 19th October 2025